

Energy Consumption Forecasting Platform

Time-Series Machine Learning Capstone Project Report

1. Executive Summary

This project presents an enterprise-grade time-series electrical energy forecasting system. The system standardizes heterogeneous multi-format energy readings (CSV, Excel, TXT, IoT streams), implements physics-aware 15-minute aggregation, and benchmarks regularized Linear Regression (Ridge), Gradient Boosted Trees (XGBoost), and Deep Recurrent LSTM Networks. Models are trained using strict chronological validation, avoiding future data leakage. Forecast horizons range from 15 minutes to 7 days with 95% empirical confidence intervals, delivered via a FastAPI backend and React 18 interactive dashboard.

2. Key Empirical Results

- Linear Regression Baseline:	MAE: 0.025 kW RMSE: 0.035 kW MAPE: 3.6% R2: 0.998
- XGBoost Regressor (Recommended):	MAE: 0.020 kW RMSE: 0.032 kW MAPE: 2.6% R2: 0.998
- Deep Recurrent LSTM Network:	MAE: 0.725 kW RMSE: 0.805 kW MAPE: 13.0% R2: 0.912

3. Physical Target Derivation

The target variable 'energy_consumption_kwh' is derived as $\text{mean}(\text{active_power_kw}) * \text{delta_t}$ (0.25h for 15-minute). Submetering active energies in watt-hours are aggregated and converted to kWh, preserving conservation of energy.